
EE 514: Robust Control Systems

Course Code	EE 514
Course Name	Robust Control Systems
L-T-P-C	3-0-0-3
Prerequisites	UG Control theory
Intended for	UG/MS/PhD
Distribution	Elective Semester: Odd/Even
Approval	9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum
Revised	11th Senate (Oct 2016) - renamed from “Introduction to Feedback Controls”

Course Contents

- **Mathematical Preliminaries:** Vector and matrix norms, Signal and system norms, Singular value decomposition, LMIs. [7 Lectures]
- **System Analysis:** System representation, sensitivity and complementary sensitivity function, concept of poles and zeros, pole and zero directions, performance limitations. [7 Lectures]
- **Feedback interconnection & Stability theory:** Well-posedness, Internal stability of feedback system, Nyquist plot, Small gain theorem. [5 Lectures]
- **Uncertainty and robustness:** Uncertainty representation (structured/parametric and unstructured), robust stability and robust performance, Kharitonov's theorem, linear fractional transformation (LFT), applications of robust control in physical systems. [12 Lectures]
- **Controller Synthesis:** Stabilizing controllers, H_∞ control, μ Synthesis. [7 Lectures]

Text Books

1. S. Skogestad and I. Postlethwaite, **Multivariable Feedback Control: Analysis and Design**, 2nd Edition, John Wiley & Sons, 2001.
2. S.P. Bhattacharyya, H. Chapellat, L.H. Keel, **Robust Control: The parametric approach**, Prentice Hall, 1995.
3. K. Zhou, J.C. Doyle, K. Glover, **Robust and Optimal Control**, Prentice Hall, 1996.
4. O.H. Bosgra, H. Kwakernaak, G. Meinsma, **Design Methods for Control Systems**, Notes for a course of the Dutch Institute of Systems and Control, Winter term 2007- 2008.
5. S. Boyd, L. E. Ghaoui, **Linear Matrix Inequalities in System and Control Theory**, SIAM, 1994.